

BeyonMLOps agentOps for enterprise IA

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SUMMARY KEYWORDS

Agent Ops, Red Hat AI, agent identity, cryptographic identity, zero trust, observability, scalability, production readiness, AI governance, multi-agent architecture, inference spikes, agent execution, security, lifecycle management, AI gateway.



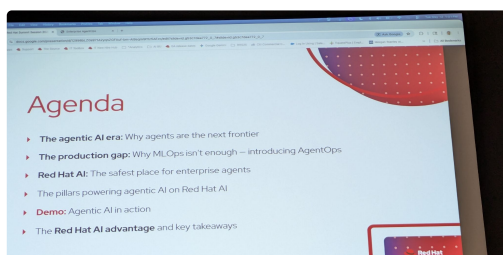
SPEAKERS

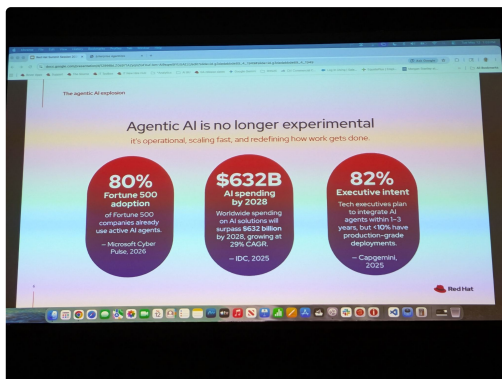
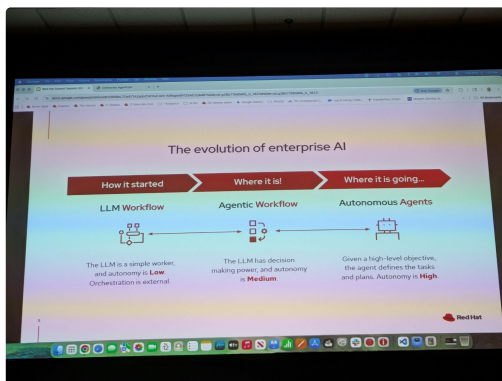
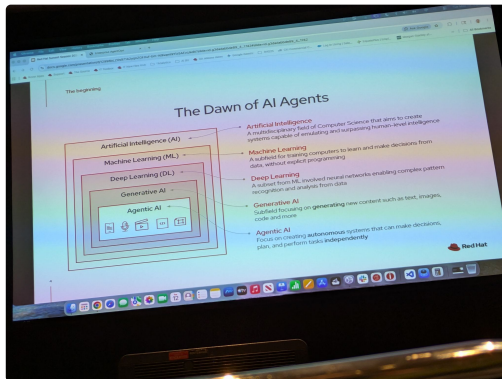
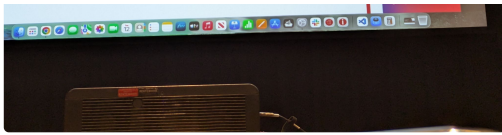
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Speaker 1 00:00

Be talking to you guys about establishing agent ops, where your enterprise Red Red Hat AI. Let's go through the agenda real quick. So we'll talk real quick in terms of, how did we achieve agentic AI, how do we get here the production gap? Why are enterprises not deploying multi agents in production? What is stopping them, and how does Red Hat AI address that, to make it safer to deploy these agents. Then we'll move on to a demo. We'll wrap it up in terms of the summary of Red Hat AI advantages, some key takeaways, if you guys don't mind, we definitely left some time at the end for Q and A so if you get it on Monday to the end, and even if we don't have time, more than happy to meet you guys outside or at the booth. We'll continue conversation. All right, so from chat box to agents. So if you guys look at this timeline, it shows that the evolution of AI as a discipline has moved from broad AI concepts to machine learning to deep learning, and then generative AI, each step was building on the last one, but the Shift Agent is different, and can just agree. So generative AI, as we know, generates content, so think text, images and code. Agentic AI, on the other hand, acts so it creates autonomous systems that make decisions, plan multi step workflows and execute tasks. And that distinction from generating to acting is what actually makes this book transformative, but also challenging for the investors. So when your AI is just answering questions that blast radius, if you will, it's kind of limited, but when autonomous systems are executing actions against production systems, the stakes are fundamentally, as we see here, to try to make it a little bit more complete. On the left, you see where more organizations started. It is pretty much closed. So the model simple order receives the front and produces an output, and all the orchestration happens external. Externally, the autonomy here is low and the risk is manageable. In the middle, we have a gentle workforce, so the LLM Valve has decision making powers. You can choose which tool to call, how to route that task. And this is where we see where the autonomy is medium. And finally, on the right side is where the market is too much at full autonomous agents. It's given a high level objective, the agent defines its own tasks and plans it. Here is where the autonomy is high. So each step to the right delivers more business value, but also demands more from infrastructure in terms of security governance, but also observability. Most organizations today are somewhere between the first and second step, first and second stage, stage, and it's all happening right now at a normal scale. These are some numbers that we got from analyst reports, where we see that 80% of Fortune 500 companies have already adopted some form of AI agent capability. IVC projects that \$632 billion in AI is by 2028 and 82% of executives intend to integrate AI agents into their operations. So we see that the adoption velocity is staggering. But here's the critical, authentic insight that we're all here and we want to learn about, is the adoption is not the same as Production Readiness. You can play around with all the open call, but no one's giving it their credit card. So most of these organizations are running pilots, proofs of concepts, or limited departmental deployments. And the gap between we're using agents and we're running agents of production. Full governance is very big, and that gap is, I think, where the real conversation is, and Peter will take us through.



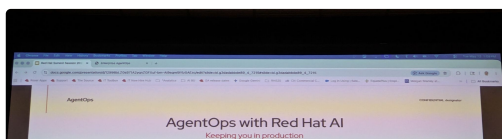
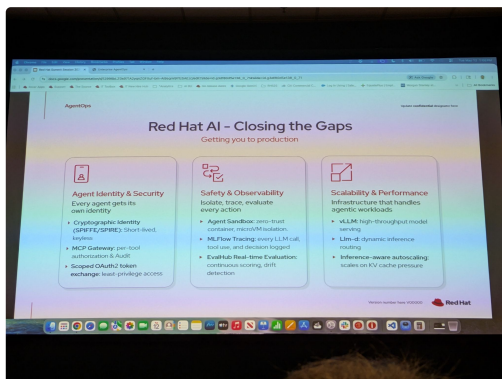
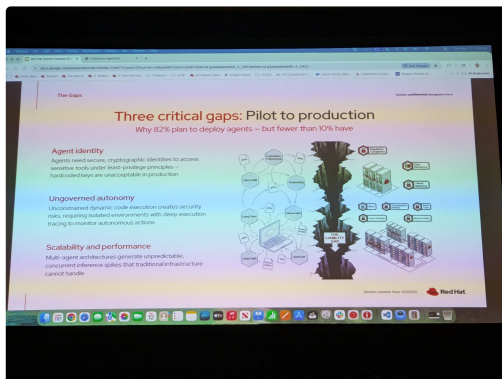


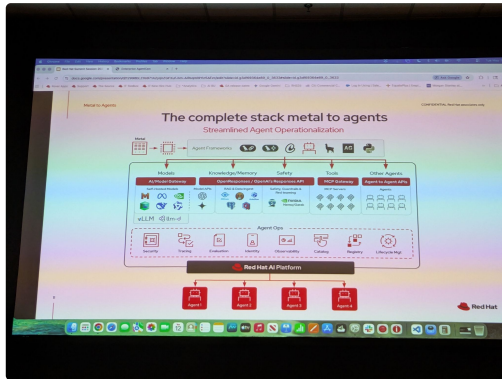
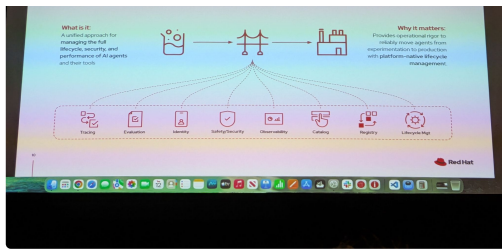
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So let's just jump into this here. So you have a proof of concept the agent works in your lab, but when you actually try to take that to production, that's where you run into the same gaps that every enterprise faces the first agent identity, when your pilot, you hard code APIs, you give broad access, but In production, that's just a security nightmare waiting to happen. So what you need is every single agent needs its own individual cryptographic identity scoped with least

privilege access the same kind of rigor that you would give to any human user. Second we have ungoverned autonomy agents. They execute code, call APIs, make decisions, and they're doing it quickly, without isolation and deep tracing of every action. You can't explain what happened when something goes wrong in a regulated industry, just saying the agent decided to do this isn't going to apply. Third up, we have scalability and performance one agent calling one model. That's easy, but when you have many agents calling many models concurrently, you're going to have inference spikes that maybe your infrastructure can't handle. You so then this is where agent ops really comes into this. And Red Hat AI is actually filling these gaps within agent identity and security. Every agent does get his own cryptographic identity. We use a spiffy spire solution, which is short lived. Keyless, auto rotated, rotating, sorry, auto rotated, auto rotated, sorry. Keys with ungoverned autonomy, we actually isolate the agent execution and zero trust sandboxes, container and micro VM boundaries, the constrain and what an entire agent can absolutely reach, really constraining that and keeping it so that way it doesn't get to information it shouldn't or try to enact any action that it should not. With each one of those actions traced, you have the ability to understand exactly what that agent has done, what it was thinking, any action that it has taken, and have deep logging and observability with that, on top of that, real time evaluation continuously scores against agent behavior so you can catch drift before that becomes a problem. Within scalability, vllm delivers high performance inference. LLMD dynamically routes inference requests across your entire fleet, and auto scaling responds to actual KV cash pressure, not just CPU. The infrastructure adapts to agentic workloads, not the other way around. Closing these gaps gets you to production, but staying in production takes even more. Here you'll see the larger picture. What we just talked about is just a piece of that puzzle. We have many other pieces, as you can see here, tracing, evaluation, identity, safety, security, observability, catalog, registry and Lifecycle Management, which we consider governance. This is for all of your AI assets, not just models, but looking through at MCPs, skills, agents, all of these pieces need to be completely governed. So you know what you're evaluating against, what you're what you're actually putting out there, versioning wise and lifecycle now this area kind of breaks down into three pillars. You have those, observe, trace, evaluate, making sure you can absolutely see what is happening second you're enforcing that identity that we just talked about before, and making sure that you have the security around these pieces, as well as proper access for each one of these ideas, one of these assets, third, develop and deploy reliably. This covers the whole governance area that I was mentioning, making sure you have what your AI, engineers and agents know that they can have access to catalogs, being able to actually see the storefronts and know what you've been given access to as an engineer, registries to contain all of that information and govern those different versions. Know what has been deployed, where it's been deployed, and how to access it, and also the ability to have safety and guardrails around these pieces red teaming, and also direct guardrails for the inference as well as for MCP and other areas that will have their own guardrails as well. The bottom line here is we're providing enterprises the optional on the operational rigor to reliably move agents from experimentation to production without locking you into a single framework. I think this gives a better look at the entire picture of the platform. Red Hat AI brings you all of the pieces you need to go from pilot to production, starting your metal, your hardware, choose the agent framework that your company wants. We don't lock you in. We are agnostic in that fact, though, we do provide actual quick starts and solutions on how to get started with your frameworks as well as bring your own. And we do have as part of the platform an opinionated framework on how to make that happen. Now you also get through that inference through models with bllm and LLMD, as well as external models. Through models as a service, you are able to contain and constrain what people have access to, but also access things like GPT, Gemini, Claude, as well as run your own models locally. But you know what you're using now because of models as a service, you know what your people are actually accessing, chargeback, cost management, all of that with self service API, to be

able to actually get API keys through self service and be able to know what you can have access to, as well as knowing what your people are actually looking to run and how much token users they have, knowledge, memory, being able to access different rag sources, different data sources, directly through an open responses, open open AI responses, along with safety through guardrails and red teaming with Nemo Garrick and Trustee AI, we have an MCP gateway, which is through quadrant upstream, and it provides the ability to make sure that your agents only have access to the tools that they should have, and not only that, through this vivis Fire solution, making sure that they are secure with token exchange, so that way you know who's accessing What and that they have the rights to do. So also agent standard agent to agent API's, so that way your your agents can communicate with other agents across the platform, as well as outside of them. And then, of course, this all sits on top of agent ops, the observability to know exactly what is happening across your entire platform, but also making sure that you have that security and all of the different governance necessities around these things. Then beneath you'll see your agents on OpenShift AI, the ability to deploy, maintain, manage all of that, observe all of your agents and know exactly what they're doing, and you can do this all without having to leave the confines of Red Hat. Now what we're going to get into now is a little bit of a demo to show through exactly how some of these pieces



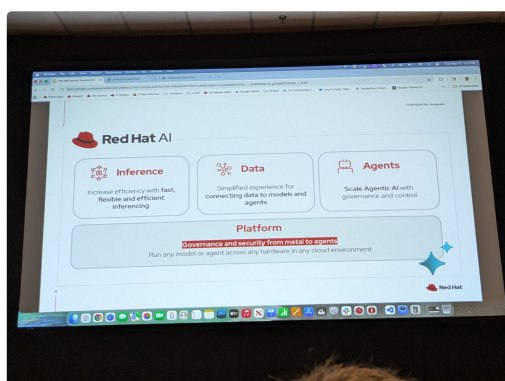


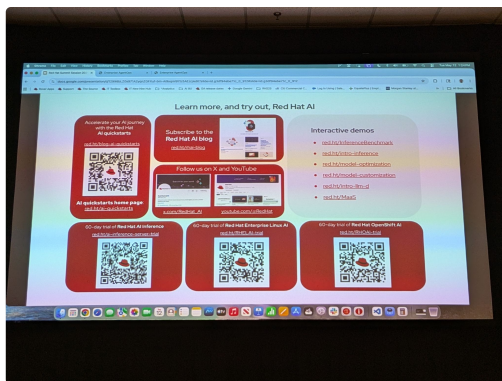
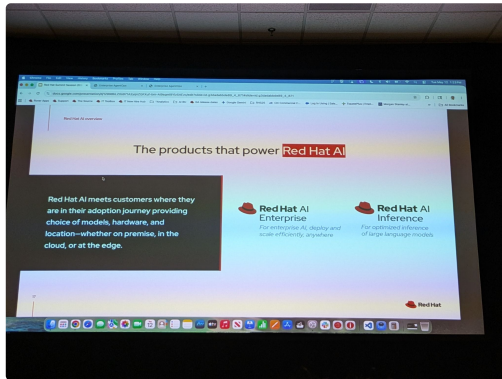
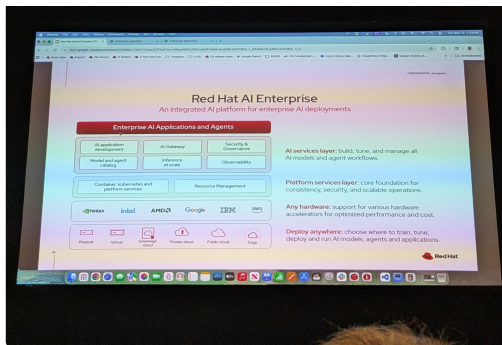
S Speaker 1 12:15
work.

S Speaker 3 12:18
So we won't see these in action with this agent ops demo, you can scan these and you will launch into the demo that we will show, essentially This demo, hopefully I can see them this demo, I and I can note right from here, you will see all right, this demo starts with this amazing Fedora capital mediator system that we built, but we have a problem because this system, and this application is only running on development, and we have our application that is running only in this developer laptop, and we wanted to go from the development to production. So we wanted to go from the inner loop into the outer loop. And we detected up a couple of things around gaps. And we detected that we have, for example, and we need to add security. We need to add also observability and the things that also Peter mentioned before. So the first thing that we want is to import our agent, and we have one piece that is fully open source, that is called caciente, and we have a principle that you can bring your own agent, so no matter how it's built, no matter how the agent framework that you're using, like, I don't know, Lang chain, crew, AI, or others, also things like open class, Well, you can import your agent, and you can enable certain things around security, also the agent to agent or the different MCPS protocols. So now we have all of these agents, but we need to be able of having this multi, multi agent architecture. So the agenda will also enable agent cards and Agent skills in order to discover the agent, from one agent to another agent. In order to start working now that we have our agent, we wanted to add, as Peter said, unique cryptographic identity. So instead of using static credentials, like if the agent gets hard, for example, or gets compromised, and we need to be able to have identity for every single agent out there. So we are using this agent deployment identity with spirit and spiffer, and using outreach, and just with one click, you can also enable the security and the workload identity across the entire site. Now we wanted to

also run the agents full secured in Sandbox. Sandboxing is trying to isolate and only allow our agent to talk to the different APIs or to the different tools that we want, and we are using the sandbox in order to prevent that. For example, the agent goes to places or calls to the API's that they are not supposed to. So we want to put these boundaries in order to control what they can access what they can also retrieve. Also we already mentioned the MCP gateway, so we inside of the banking application. We need some tools in order to receive some data actions against databases and so on, so forth. But we wanted to have control on the different MCP servers that we want. So instead of relying on this will not be harm. We wanted to maintain the control and the governance of the different MCP servers, and we are putting the MCP gateway in front of in order to have access control and also observability end to end, so we know exactly that we can allow our agent to access only certain tools as well. Then we have most as a service. An agent without a model is useful because you need to power the AI agent with modes as a service in order to have this flexibility. And we are using this in order to have this private AI in France, in order to power our AI engines, so we can have these different models that we see, and also we can import here once if, for example, we wanted to run or access to certain inputs of the models that are not running in the same cluster. So this is super flexible, and you can give this end to end observability as well. Then talking about observability, you need to know when things go wrong, **what's happening. So I went that road. And when you are trying to debug a multi agent system, you don't know exactly when you start to look in. And this is giving** you an end to end observability. So you will see in here all of the tool coding, all of the chat completion as well. So all of these AI model calls, and you will have an end to end traces that you can you can do things like audits and all type of things that are super helpful, not only for troubleshooting, also to do this end to end audit that, for example, our bank relies and needs. On the other hand, we need to ensure that the AI agent not produce things like very harmful or produces things that could be also bad. So we are providing guardrails and safety using Nemo guardrails and using also FMS orchestration in order to have these guardrails and put some safety in front of the agent. So you will see in here how we can prevent things that are harmful or, for example, biased or are not very good, so they will prevent it. And last month of this, we want to have this popular identity. So this is also one example around having, for example, splitted and having one identity per agent, so we can prevent, like the agent access all the things that they are not supposed to do, and also having this end to end worker identity, we can identify every single step in the End to End tracing, for example, and make this isolation for the different agents in order to keep the agents in their own scope, in order that you also prevent bad things happening. So all of these comes together, and we already explained this, but essentially, we are trying to have this philosophy of bring your own agent, and we will make a production ready, adding all of these layers that we already mentioned, capwelling, safety agent, life cycle and so on, so forth. So coming back again to, and I don't know, all right, coming back to the the couple of things around the rest of the AI advantage. We have these by taking control of the AI environment, the organization also can have this control. So that's exactly that we are getting with Western AI. Western AI is our enterprise platform. You already see this. So we have inference, we have data, and we have also the agent. And everything is running on a platform with these governance and security that we call from metal to agents, because you can run all the stack fully secure, and also you can run any model or any agent across any hardware and any cloud environment. So you already also see in the keynote this morning. And the good thing is that it's deployable everyone. So you can also deploy this in your premise. You can have this hybrid solution that our customers are doing as well. So having, for example, some Indian premise on the bare metal and going also to the cloud and rely on this, because always this is installing on top of OpenShift and also on top of Red Hat. So to this, we've organized our AI strategy in these four pillars, again, the imprints, the data, the agents, and running on top of the platform and that we already saw. We have this rest of the price. We have the different layers that starting, for example, from the bottom. We have the

principle of deploying anywhere, so you can deploy in any scenario, like bare metal or going to the happy cloud as well. You can run in any accelerator. So supports various hardware accelerators, as we already saw, you can find it in video, but also in Intel, AMD, Google and in other accelerators, we have this platform service layer that we have the confirmations, and then we have the resource management and the other things. And then on top of the AI services layer, that you can build, tune and manage all of the AI models and also agent workflows. And on top of that is like the enterprise AI application and agents that we can run. So with that, we have this portfolio that is designed to meet customers where they are, in the out and learning. We have Red Hat Enterprise, AI enterprise that is integrated AI platform for enterprises looking to deploy and scale AI models, agents and applications efficiently and anywhere, also resident AI in frame service that these are product for enterprises that they want to optimize their inference, and they want to also deploy these ILMs, this last year was models, and afterwards, we have open CPI that enables enterprise to extend the capabilities to OpenShift and also support AI. And last but not least, green AI provides out of the box platform to consistently running an LMS into a individual in several environments. And these final slides contains a set of links and materials that we have covered. You can check that out, and you can also have several interactive demos. You can access to this, and some of them will be published very soon. So all of these. You have it online and available, so take a look. We will also publish the PDFs on these slides, and I think that that's all for this session. Thanks for attending. And I don't know if you have any questions, but we will keep





S Speaker 4 24:02
 this running. I forgive me if I seem confused. Is agent ops a separate thing, or is that a label on what we're calling all of this?

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Speaker 2 24:11

So the agent ops itself is, if we go back you so agent ops itself is actually the pieces that help make the actual agents operationalized. It's not the actual prompting, it's not any of those pieces, but it is the governance of those things, it is actually observing them and making sure you have full tracing down the stack. It's the evaluation of them. It's the things that are going to make that work. But think of it as day two, and really making sure what you've built in the labs and what you build in your proof of concepts can actually be in production, and checking for drift, checking for any issues that may arise, remediation of those issues. That's the part of that is Agent ops itself. Everything else is actually, you know, just the actual AI pieces. And agent ops makes sure that keeps an eye on each one of those to know what your what you have in production. Where is it in production? How, who's using it, all of those pieces. So that way you can intelligently decide this is working, this is scaling properly, and the number of users around it, you know, who's using what? When you can deprecate things, when you can move things forward, and helping lifecycle, all those areas.

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Speaker 4 25:35

Thank you

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Speaker 2 25:35

absolutely. Yes. So and those are continually being built out as well. So to be clear with everybody, on the roadmap at this time, MCP is really leading the way. It's not the shining star we all thought it was a year ago in terms of exactly an MCP everything situation, but it's leading the way in showing us what we need in terms of catalogs, registries, lifecycling, making sure that we can keep eyes on all of that. Skills and agents are really settling right now in terms of letting the dust settle, figuring out how we want to make sure that those pieces are implemented. So we're going to have skills registry. We're releasing three four this week. Three five is in August at this point, and in three five, skills will have a full registry, sorry, skills will have full catalog. Agent, catalog capabilities. We're working in if any of you have heard of ml flow, the ML flow upstream is actually in GA and three four, and we're working with data bricks directly in the upstream to actually bring a registry capability for those solutions. So that one is in the works right now, with data works, it's going to follow the exact same methods, though, as MCP. So we're going to have cattle registry, actual life cycle, actual lifecycle of those in three five though we also have are working with K gente upstream. K Gen T is for the agent part of that so deployment lifecycle, and being able to show deployments within within OpenShift, and making sure those are available as well. All of these are in development, and making sure that we're increasing those capabilities, but working directly with the upstream to make sure that we all land on something that will be more long lived, and not just trying to get something out the door as fast as possible. So that idea is going to keep expanding. Now, guardrails, knowledge, sources, pipelines, notebooks, there's so many agent or sorry, AI, assets and all of that needs to be governed. So that way, when you know something's gone wrong, you can actually check to know what version it was. Have Supply Chain Management around that. Make sure that you aren't just bringing in things, which I guarantee everybody in this room is dealing with that at the moment, in terms of AI engineers throwing up an MCP server, npm install, opening up some pretty crazy security gaps within your system. So this is all about you being able to have a trusted environment and making sure that you, if I could say anything, this is the gear of trust. You can't go to production without that trust. And yes, you can get something out there really quickly, but I wouldn't trust any of that, so I would be very careful with with that as well. Yes, sir. So right now we have two gateway solutions, and we are already looking at bringing them under one roof. Inference came out last year. You're all you know in terms of being able to run production workloads with inference. So we absolutely immediately made what people were calling the AI gateway, but it's really an inference gateway. Then the MCP gateway was there to make sure that we could have a proper security solution. That's where spiffy spire first got introduced to this, to make sure that we can handle zero trust. So make having that gateway in there. Now we're looking at what the best solution, and we're not just picking one and going with it. We're going to have an AI gateway that can cover all of your different workloads coming through it. But we want to have an AI native proxy. We want to make sure that when you're actually looking at the packets that is for AI and that it doesn't copy packets and move them so we are looking at what our solution is going to be that is actively being looked at and developed right now. And yes, everyone's got an AI gateway now. There's 100 of them out there. We don't just want to pick one, because it's the thing we're trying to make this a long term solution that will be there to actually have a rapid, efficient proxy with proper packet, well, packet analysis without having to copy that out, which is two of the main pieces. But yes, there will be a full AI gateway umbrella for all AI workloads, but the MCP gateway and inference gateway are two main that can help you get to production at this moment. Yes, sir. Can

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Speaker 5 29:55

you talk more about the token exchange between two agents, like what kind of protocols are being used and how the security is

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Speaker 3 30:06

handled? Yeah, that's a very good question. So when two agents, for example, every single agent, have their own cryptography, identity, so we are using things like spiritual prefer that we know that. It's not known, but it's giving specifically workload identity and workload name in order to identify specifically these, these and these. And when you are trying to also communicate to another agent, we have outreach that will inject one sidecar and will detect when this is valid or not. So if you are trying to hijack will be intercepted, will be validated and will be discarded, so will be automatically. And you don't need to do anything. It's like with Istio, with the different side cards. You don't need to change your agent code. It's just enabling out the outreach, and it's just one click, automatically will inject this and will identify your agent that is communicating to another agent or to another tool as well. So all of these end to end will have these. And if you see, for example, into the demo, we will see these. So I'm switching back to the demo, and in this demo, you are seeing, for example, the fourth bridge. In here. Check this observability. You will see also the outreach. And we can also see this into the traces. So you can have this end to end observability having inside of the own traces. So in this trace, for example, we will see, top right, you have this PFA, and this is identifying specifically that this workload is having this specific identity. Is

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Speaker 5 31:50

there any kind of stress load testing happening here, or is there

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Speaker 3 31:58

any queuing happening when multiple wants to connect to Yeah, it's, we did some testing, and it's super, super lean, so we are not adding a lot of like steps. The only thing that is happening is going to a sidecar that will have the exact same net namespace, so the latency will be minimal. And this is super like, not resource hungry. No will be happening one by one, and you have some certain queuing, but it's super fast, because the only thing that is doing is like one another another. So it's some sort of queuing, but inside of the outfit as

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Speaker 5 32:38

well,

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Speaker 3 32:40

correct? It's a sidecar. So we'll go throughout the sidecar automatically. We go to the endpoint, and we'll have this validation, yeah, so building on, like the resource

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Speaker 6 32:52

question, so in some environments, like thinking about how we deploy, we may have limitations as to how many models we could have that models we could have any sort of paradigm for spinning out models dynamically

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Speaker 3 33:09

on demand. We have a talk, in fact, tomorrow that is called dynamic dynamic workloads, in order to have, like, more models than GPUs, and in order, in order to load automatically, one thing though, that we need to take into consideration is that the model loads is usually taking a little bit because these models are very adjunct and sometimes you need to wait a little bit until this happens, you can have some warm up GPUs in order to scale up, and we have some dynamic auto scaling using Keta. And also another thing that we will I'm not doing any spoiler, because I want to you go to the session as well, but itself is like an interceptor analogy, scale up automatically, but yeah, it's sometimes a little bit tricky to manage. This earlier

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Speaker 7 34:08

you mentioned trust, and I presume the AW can use the skills. So how can we make sure that those skills out of trust? So

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Speaker 2 34:24

thank you. That's a wonderful question, and that is exactly why agent Ops is so important, is that it is creating pipelines in of these for your company to validate. We are getting all of the pieces in place for that governance. Part of the lifecycle as well, is going to be pipelines directly in to allow you to verify scan, make sure that these skills are coming from the trusted sources, and then making sure, then that you can keep them quarantined in a state, that they can be checked and make sure that they're okay once they've been checked, then you can publish them out to your catalog, make sure that they're going to be loaded into the registry correctly, and that is why that is exactly so important right. Now, many of you probably just have skills being transferred across your organization. Oh, you have a good skill. Send me that real quick internally. That's not so bad, but when you start pulling skills from outside, that's going to be a major problem, and there's a lot of skills that they look wonderful, and there's that one little code snippet that you missed that just opened up your entire network. So it is incredibly important that you have this governance solution. When you're first developing an agent, people don't see that importance, because they just want to get the agent up and running. They just want to make sure it's there. But when you want to go to production, you have to make sure your supply chain is checked. You have signing security, all of the different certification around those and as platform engineers, as the enterprise itself, you want to have a marketplace where trusted skills can be surfaced that aren't just GitHub or out there in the wild or sent through email to people, but yeah, that is the key there is that you want to make sure that you have a pipeline in place with approvals to say that these should go out and these are enterprise approved evaluation.